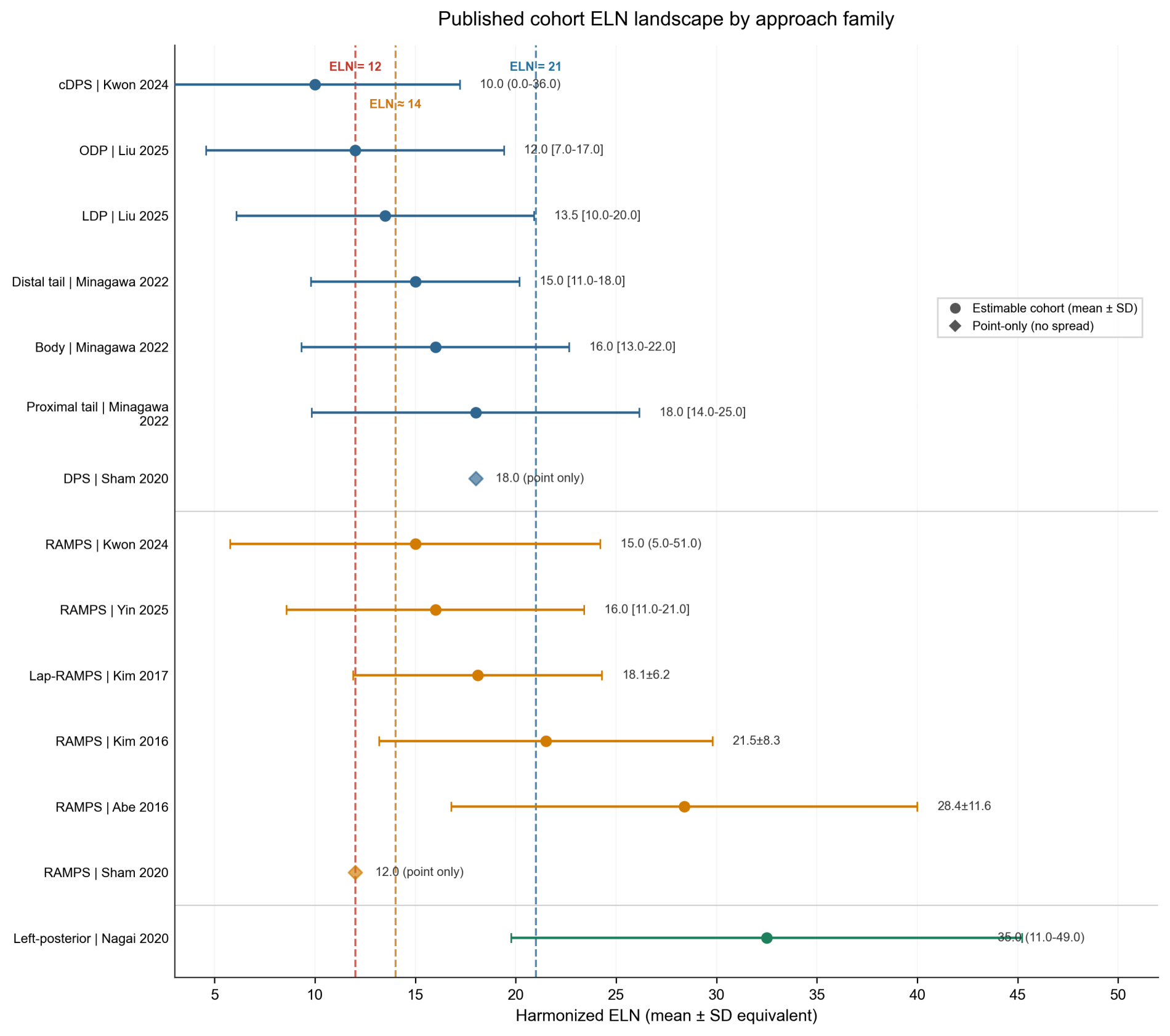

Published cohort ELN landscape by approach family
ELN= 12
ELN≈ 14
ELN= 21
cDPS | Kwon 2024
10.0 (0.0-36.0)
ODP | Liu 2025
12.0 [7.0-17.0]
LDP | Liu 2025
13.5 [10.0-20.0]
Distal tail | Minagawa 2022
15.0 [11.0-18.0]
Body | Minagawa 2022
16.0 [13.0-22.0]
Proximal tail | Minagawa 2022
18.0 [14.0-25.0]
DPS | Sham 2020
18.0 (point only)
RAMPS | Kwon 2024
15.0 (5.0-51.0)
RAMPS | Yin 2025
16.0 [11.0-21.0]
Lap-RAMPS | Kim 2017
18.1±6.2
RAMPS | Kim 2016
21.5±8.3
RAMPS | Abe 2016
28.4±11.6
RAMPS | Sham 2020
12.0 (point only)
Left-posterior | Nagai 2020
35.0 (11.0-49.0)
Estimable cohort (mean ± SD)
Point-only (no spread)
5
10
15
20
25
30
35
40
45
50
Harmonized ELN (mean ± SD equivalent)